Unramified algebras over a discrete field

A constructive product decomposition

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Abstract

We prove that a finitely presented, formally unramified algebra over a discrete field is a finite product of separable monogenic algebras. The diagonal idempotent supplies flatness transfer, regularity and trace duality. A bounded search then gives either a global primitive element, obtained from a tensor separator, or a finite coefficient field over which an explicit idempotent splitting produces the required factors. The proof uses finite linear algebra and finite searches.

Part I
The mathematical argument

1 Statement and conventions

All rings are commutative with identity, and algebra maps preserve identity. The base k is a nonzero *discrete field*: equality in k is decidable and every nonzero element has an inverse. Algebras over k may be zero. A module is *finite* if it has a finite spanning family; a *finite basis* includes unique coordinates. We distinguish these two notions throughout the proof. An *explicitly finite* set comes with a finite enumeration and decidable equality; this is stronger than finite generation of a module.

Definition 1.1. An R-algebra C is *formally unramified* if, for every R-algebra T and ideal N with $N^2 = 0$, the map

$$\mathrm{Hom}_{R\text{-alg}}(C, T) \longrightarrow \mathrm{Hom}_{R\text{-alg}}(C, T/N)$$

is injective. A monic polynomial $f \in k[T]$ is *separable* if f' is invertible in $k[T]/(f)$, equivalently if $uf + vf' = 1$ for some $u, v \in k[T]$.

Theorem 1.2. *Let C be a finitely presented, formally unramified k-algebra. There exist finitely many monic separable polynomials $f_i \in k[T]$ and a k-algebra isomorphism*

$$C \simeq \prod_i k[T]/(f_i).$$

In particular, C has a finite k-basis and is finite étale over k.

The factors in this statement are monogenic étale algebras. Irreducibility of the f_i is not required. The polynomial 1 is allowed: its quotient is the zero algebra, and its derivative is a unit in that ring. Such factors can be omitted when forming the product.

We first construct the diagonal idempotent and deduce regularity. An elementary finiteness argument then provides a basis and trace duality. The final step has two branches, chosen by a bounded finite calculation; it requires no decision whether k is finite or infinite.

2 The diagonal idempotent, flatness and regularity

Write

$$D = C \otimes_R C, \qquad \mu : D \longrightarrow C, \quad \mu(a \otimes b) = ab, \qquad \Delta a = a \otimes 1 - 1 \otimes a.$$

Lemma 2.1. *If C is a finitely generated, formally unramified R-algebra, there is an element $\varepsilon \in D$ such that*

$$\mu(\varepsilon) = 1, \qquad \varepsilon \Delta a = 0 \quad (a \in C), \qquad \varepsilon^2 = \varepsilon.$$

Moreover $\ker \mu = (1 - \varepsilon)D$.

Proof. Put $J = \ker \mu$. If $x_1, \ldots, x_r$ generate C as an algebra, then J is generated by $\Delta x_1, \ldots, \Delta x_r$. Indeed, after these differences have been killed, the two maps from C coincide, and multiplication identifies the resulting quotient with C.

The maps $C \rightrightarrows D/J^2$ agree after passage to D/J. The kernel J/J^2 is square-zero, so formal unramifiedness makes the maps equal in D/J^2. Hence $J = J^2$.

Here is the determinant argument for this finite idempotent ideal. Choose generators $u_1, \ldots, u_r$ of J and write $u_i = \sum_j a_{ij} u_j$ with $a_{ij} \in J$. The adjugate identity for $I - (a_{ij})$ shows that

$$\varepsilon = \det(I - (a_{ij}))$$

annihilates every u_i. Also $\varepsilon \equiv 1 \pmod{J}$. Thus $\varepsilon J = 0$, $1 - \varepsilon \in J$, and $\varepsilon(1 - \varepsilon) = 0$. For $u \in J$, the equality $u = (1 - \varepsilon)u$ gives $J = (1 - \varepsilon)D$. The remaining assertions follow immediately. □

Lemma 2.2 (Flatness transfer)**.** *Suppose $R \to C$ has the diagonal element of Lemma 2.1. If a C-algebra T is flat over R, it is flat over C.*

Proof. The C-module $C \otimes_R T$ is flat by base change. Let ε_T be the image of ε in this algebra. Multiplication $C \otimes_R T \to T$ has the section

$$t \longmapsto \varepsilon_T(1 \otimes t).$$

The equality $\varepsilon_T(c \otimes 1) = \varepsilon_T(1 \otimes c)$ makes this section C-linear, and $\mu_T(\varepsilon_T) = 1$ proves the section equation. A module retract of a flat module is flat. □

Proposition 2.3. *A finitely generated, formally unramified algebra C over k is von Neumann regular:*

$$\textit{for every } a \in C \textit{ there is } b \in C \textit{ with } a = a^2 b.$$

Consequently C is reduced.

Proof. Every module over a discrete field is flat. The equational criterion proves this directly: in a finite relation $\sum_i \alpha_i t_i = 0$, either all coefficients vanish or one coefficient α_p is invertible. In the second case, the vectors $e_i - \alpha_i \alpha_p^{-1} e_p$ generate the relations needed by that criterion. This argument requires no choice of a basis of the module.

Apply Lemma 2.2 to $C/(a)$. This quotient is flat over C. Tensoring the injection $(a) \hookrightarrow C$ with $C/(a)$ therefore gives an injection

$$(a)/(a)^2 \longrightarrow C/(a).$$

The map is zero, so $(a) = (a)^2$ and $a = a^2 b$ for some b. Iterating this equation gives $a = a^m b^{m-1}$ for $m \geq 1$. Thus every nilpotent element is zero. □

3 Finite spanning families, bases and minimal polynomials

Definition 3.1. Call a ring *algebraically zero-dimensional* if

$$\forall x\ \exists N \geq 0,\ b \qquad x^N(1 - xb) = 0.$$

Regular rings have this property with $N = 1$. We use the displayed equation itself, without any definition through prime ideals.

Lemma 3.2 (Integral descent)**.** *If $A \hookrightarrow B$ is integral and B is algebraically zero-dimensional, then so is A.*

Proof. For $x \in A$, choose N, b with $x^N = bx^{N+1}$ in B, and a monic equation

$$b^h + c_{h-1}b^{h-1} + \cdots + c_0 = 0, \qquad c_i \in A.$$

Induction gives $x^N = b^j x^{N+j}$. Hence, for $0 \leq j \leq h$, $b^j x^{N+h} = x^{N+h-j}$. Multiplying the monic equation by x^{N+h} yields

$$x^N(1 + c_{h-1}x + \cdots + c_0 x^h) = 0.$$

This is the required equation in A, by injectivity. A degree-zero monic equation gives $1 = 0$, when the assertion is immediate. □

Proposition 3.3. *A finitely generated, algebraically zero-dimensional k-algebra is finite as a k-module. If it is finitely presented as an algebra, it has a finite k-basis.*

Proof. For the first assertion, induct on the number of algebra generators $x_1, \ldots, x_r$. With no generators, 1 spans the algebra. For $r > 0$, choose N, b with $x_1^N(1 - x_1 b) = 0$ and write $b = P(x_1, \ldots, x_r)$. The polynomial

$$F = X_1^N(1 - X_1 P)$$

vanishes at the generators and is nonzero: its coefficient at $X_1^N X_2^0 \cdots X_r^0$ is 1.

Choose $M \geq 2$ larger than all exponents occurring in F. Make the invertible triangular substitution

$$X_i = Y_i + T^{M^i} \quad (1 \leq i < r), \qquad X_r = T.$$

The top T-degree of a monomial X^α is $\alpha_r + \sum_{i<r} \alpha_i M^i$. These degrees are distinct, by uniqueness of base-M expansions. Consequently the highest T-coefficient of the transformed F is a nonzero scalar in k. Dividing by it gives a monic equation in T. A nonzero constant relation gives the zero algebra and can be handled immediately.

Let A' be the image subalgebra generated by the new $r-1$ variables, and let t denote the image of T. The inverse substitution gives $C = A'[t]$. If the monic equation has degree d, then $1, t, \ldots, t^{d-1}$ span C over A'. In particular C is integral over A'. Here one can use the determinant trick: multiplication by an element, written on a finite spanning family containing 1, gives a matrix whose monic characteristic polynomial annihilates that element. Lemma 3.2 makes A' zero-dimensional. Induction makes A' finite over k, and combining the two spanning families proves the assertion for C.

For the basis assertion, write $C = k[X_1, \ldots, X_r]/(g_1, \ldots, g_s)$. The determinant trick and the finite spanning family give monic annihilators $p_i(X_i)$ of the original generators. The algebra

$$Q_0 = k[X_1, \ldots, X_r]/(p_1(X_1), \ldots, p_r(X_r))$$

has its explicit bounded-monomial basis. Its quotient by the images of the g_j is C. The ideal generated by these images is spanned over k by their products with the bounded basis monomials of Q_0. Gaussian elimination on this finite list of vectors constructs a basis of C. This step uses the original finite presentation; it does not require a finite presentation of the intermediate image subalgebras. □

Lemma 3.4. *A reduced k-algebra which is finite as a k-module is regular.*

Proof. For a choose a monic annihilator $p(T)$ by the determinant trick. Decidable equality in k allows us to write $p(T) = T^s(c + Th(T))$ with $c \neq 0$. If $s = 0$, then $c + ah(a) = 0$. If $s > 0$, the element $a(c + ah(a))$ is nilpotent and is therefore zero. In either case $ca + a^2h(a) = 0$, so $a = a^2(-c^{-1}h(a))$. □

Lemma 3.5. *Let C have a finite k-basis, and let $x \in C$. There is a monic polynomial f of least degree with $f(x) = 0$. Evaluation induces an injection $k[T]/(f) \hookrightarrow C$, whose image is $k[x]$.*

Proof. Use Gaussian elimination to test successively whether x^m is in the span of $1, x, \ldots, x^{m-1}$. A monic annihilator supplies a finite bound for this search. The first dependence gives f. There is no nonzero polynomial of smaller degree vanishing at x: divide such a polynomial by its nonzero leading coefficient to obtain an earlier dependence. Division by the monic polynomial f now shows that the kernel of evaluation is precisely (f). □

4 Trace duality and the monogenic separability criterion

For a finite-basis algebra C, write $\mathrm{Tr}(a)$ for the trace of multiplication by a.

Proposition 4.1. *Suppose C has a basis $b_1, \ldots, b_n$ and a diagonal idempotent ε as in Lemma 2.1. Write $\varepsilon = \sum_i b_i \otimes c_i$. Then*

$$a = \sum_j \mathrm{Tr}(ab_j)c_j \qquad (a \in C). \tag{1}$$

In particular the trace pairing is perfect.

Proof. Let λ_i be the basis coordinate maps. Comparing coefficients in $(z \otimes 1)\varepsilon = (1 \otimes z)\varepsilon$ gives

$$zc_i = \sum_j \lambda_i(zb_j)c_j.$$

Substitute $z = ab_i$ in the ith equation and sum over i. The left side is a because $\sum_i b_i c_i = \mu(\varepsilon) = 1$. The coefficient of c_j on the right is

$$\sum_i \lambda_i(ab_ib_j) = \sum_i \lambda_i((ab_j)b_i) = \mathrm{Tr}(ab_j),$$

which proves (1). Applying coordinate maps to this identity gives an inverse for the square matrix $(\mathrm{Tr}(b_ib_j))$. □

Lemma 4.2. *If C has a finite basis and nondegenerate trace pairing, then every idempotent factor eC has these properties.*

Proof. The image of multiplication by e has a basis obtained by Gaussian elimination. As k-modules, $C = eC \oplus (1-e)C$. Multiplication by $a \in eC$ acts as zero on the second summand, so $\mathrm{Tr}_C(a) = \mathrm{Tr}_{eC}(a)$. If $a \in eC$ pairs to zero with every element of eC, then for every $b \in C$,

$$\mathrm{Tr}_C(ab) = \mathrm{Tr}_{eC}(a(eb)) = 0.$$

Nondegeneracy on C gives $a = 0$. □

Lemma 4.3. *Let C be a reduced finite-basis k-algebra with nondegenerate trace. If $C = k[x]$ and f is the minimal monic polynomial of x, then f' is invertible in $C \simeq k[T]/(f)$.*

Proof. First, every k-derivation $\delta : C \to C$ is zero. For $a, b \in C$, write m_a for multiplication by a. The identity

$$[b\delta, m_a] = m_{b\delta(a)}$$

and vanishing of the trace of a commutator give $\mathrm{Tr}(b\delta(a)) = 0$. Nondegeneracy gives $\delta(a) = 0$.

If $f'(x)h = 0$, the assignment $x \mapsto h$ defines a derivation on $k[T]/(f)$: it sends a polynomial g to $g'(x)h$, and the relation f maps to zero. Hence $h = 0$. Finally, regularity from Lemma 3.4 supplies $f'(x) = f'(x)^2 b$. The element $1 - f'(x)b$ annihilates $f'(x)$ and must vanish. Thus $f'(x)b = 1$. □

5 Constructing a tensor separator from finitely many scalars

Fix a basis $b_1, \ldots, b_n$ of C and its diagonal idempotent $\varepsilon \in D = C \otimes_k C$. We may assume $n > 0$, since $n = 0$ gives the zero algebra. We seek $x \in C$ and $w \in D$ such that

$$(\Delta x)w = 1 - \varepsilon. \tag{2}$$

Lemma 5.1. *Let B be a finite-basis algebra over k. If the coefficients of $g \in B[Z]$ generate the unit ideal of B, multiplication by g on $B[Z]$ is injective.*

Proof. Suppose $gq = 0$ for a nonzero q. Among nonzero annihilators of degree at most $\deg q$, choose one of least degree, say q_0 of degree s. This is a finite linear-algebra construction: for each degree bound, the annihilating equation is a finite homogeneous system over k.

Write $g = \sum_{i=0}^{d} a_i Z^i$. Comparing highest coefficients shows that $a_d q_0$ has degree less than s. It is again an annihilator of g, so minimality gives $a_d q_0 = 0$. Now $\sum_{i<d} a_i Z^i$ also annihilates q_0. Descending through its coefficients gives $a_i q_0 = 0$ for every i, by the same argument. A linear combination of the a_i is 1, so $q_0 = 0$, a contradiction. □

Proposition 5.2. *If k contains $n^3 + 1$ specified distinct scalars, then (2) has a solution.*

Proof. The ideal $\ker \mu$ is generated by the Δb_i, since the b_i span C and therefore generate it as an algebra. Since $1 - \varepsilon \in \ker \mu$, the elements $\varepsilon, \Delta b_1, \ldots, \Delta b_n$ generate the unit ideal of D. Apply Lemma 5.1 to

$$g(Z) = \varepsilon + \sum_{i=1}^{n} \Delta b_i Z^i \in D[Z].$$

The algebra D has the basis $b_i \otimes b_j$, of size n^2. Thus multiplication by g is an injective square matrix over $k[Z]$. Its determinant $h(Z)$ is nonzero: otherwise Gaussian elimination over $k(Z)$, followed by clearing denominators, would give a nonzero polynomial vector in its kernel. Each matrix entry has degree at most n, so $\deg h \le n^3$.

A nonzero polynomial of degree at most n^3 cannot vanish at all the specified n^3+1 distinct scalars. Choose a with $h(a) \neq 0$. The adjugate matrix makes multiplication by $g(a)$ invertible, hence $g(a)$ is a unit of D. Put

$$x = \sum_{i=1}^{n} a^i b_i, \qquad w = (1-\varepsilon)g(a)^{-1}.$$

Since $g(a) = \varepsilon + \Delta x$, $\varepsilon \Delta x = 0$ and $\varepsilon(1-\varepsilon) = 0$, these elements satisfy (2). □

6 A tensor separator gives a global primitive element

Lemma 6.1. *An injective ring map from a regular ring reflects units.*

Proof. If $a = a^2 b$, then $1 - ab$ annihilates a. When the image of a is a unit, the image of $1 - ab$ is zero. Injectivity gives $ab = 1$. □

Proposition 6.2. *Let C be reduced with a finite k-basis and a diagonal idempotent. If x, w satisfy (2), then $C = k[x]$, and the minimal monic polynomial of x is separable.*

Proof. Let f be the minimal monic annihilator of x, of degree m. If $m = 0$, its equation is $1 = 0$, and the assertion is immediate. Assume $m > 0$. There is a polynomial $H(X, Y)$ with

$$f(X) - f(Y) = (X - Y)H(X, Y), \qquad H(Y, Y) = f'(Y).$$

It is monic of degree $m - 1$ in X. These identities follow from Horner recursion: the divided difference of $a + TF(T)$ is $F(Y) + XH_F(X, Y)$.

Evaluate at $X = x \otimes 1$, $Y = 1 \otimes x$, and denote the result by H_x. The root equation gives $(\Delta x)H_x = 0$. Multiplication by w yields $(1 - \varepsilon)H_x = 0$. Equalisation on the diagonal then gives

$$H_x = \varepsilon H_x = \varepsilon(1 \otimes f'(x)). \tag{3}$$

Minimality makes $1, x, \ldots, x^{m-1}$ linearly independent over k. Gaussian elimination constructs a linear functional $\lambda : C \to k$ with

$$\lambda(1) = \cdots = \lambda(x^{m-2}) = 0, \qquad \lambda(x^{m-1}) = 1.$$

For $m = 1$, the first list of conditions is empty. One explicit construction is to project onto the quotient by the span of the lower powers, take a nonzero coordinate of the image of x^{m-1}, and normalise it. Contract the left tensor factor of (3) by λ. Since H is monic in X, this gives

$$1 = f'(x)(\lambda \otimes 1)(\varepsilon). \tag{4}$$

Thus the derivative already has an explicit inverse in C.

Let $Q = k[T]/(f)$, viewed as the subalgebra $k[x] \subseteq C$ by Lemma 3.5. It has its monomial basis and is reduced, so it is regular by Lemma 3.4. Lemma 6.1 therefore puts $f'(x)^{-1}$ in Q. In particular f is separable over k.

It remains to show that $Q = C$. Write $\varepsilon = \sum_j b_j \otimes c_j$ in an arbitrary basis of C. Every right-hand coordinate of H_x in this left basis belongs to Q: expand the powers of x in the left tensor factor in the basis b_j; the remaining coefficients are polynomials in the right-hand x. By (3),

$$\varepsilon = H_x(1 \otimes f'(x)^{-1}),$$

so every c_j belongs to Q as well. Finally (1) expresses each $a \in C$ as $\sum_j \operatorname{Tr}(ab_j)c_j$, an element of Q. □

7 Bounded coefficient closure and the small-field branch

Lemma 7.1. *Given finitely many scalars in k and a natural number M, one can construct either $M+1$ distinct elements of k, or an explicitly finite subfield $k_0 \subseteq k$ containing all the given scalars.*

Proof. Maintain a finite list of distinct scalars. Test whether it contains $0, 1$ and the given scalars and is closed under addition, negation, multiplication and total inverse, with $0^{-1} = 0$. Each test is finite and decidable. If it fails, the failure exhibits one missing scalar, which is appended. If it succeeds, the listed elements form the required finite subfield. Stop as soon as the list has $M+1$ elements. Starting with the empty list, at most $M+1$ unsuccessful tests are possible before this alternative is reached. □

Lemma 7.2. *An explicitly finite, reduced algebra over a discrete field is a finite product of explicitly finite fields, with zero factors allowed.*

Proof. The algebra A is regular by Lemma 3.4. For each element a in a finite enumeration choose b with $a = a^2 b$. Set $f = ab$. Then

$$f^2 = f, \qquad a(1-f) = 0.$$

On the factor fA, the element fa has inverse fb; on $(1-f)A$, the image of a is zero.

Process the enumeration successively. A current factor eA is replaced by efA and $e(1-f)A$. An element already known to be zero or invertible retains that property on both new factors. After the finite list has been processed, every element of each factor is zero or a unit. Equality is decidable, so each factor can be tested for being the zero ring. Every other factor is a field. Its finite enumeration is obtained by projecting the original enumeration and deleting duplicates.

At each stage the maps

$$eA \longrightarrow efA \times e(1-f)A, \quad z \longmapsto (fz, (1-f)z), \qquad (u, v) \longmapsto u + v$$

are inverse algebra maps. Composing them proves the product assertion. Termination is induction on the enumeration length, without any search for a prime ideal. □

Lemma 7.3. *An explicitly finite field F is generated by one element over any subfield $k_0 \subseteq F$.*

Proof. The multiplicative order of each nonzero element can be found by finite search in its powers. Choose γ of maximal order m. For two commuting elements of orders r, s, there is an element of order $\operatorname{lcm}(r, s)$: factor the positive integers r, s, take for each prime a power of the element with the larger prime-power order, and multiply these components. Components of coprime orders have product of order the product, by Bézout's identity. This is a finite calculation with integers and group operations.

Maximality of m now implies that the order of every nonzero element of F divides m. All nonzero elements are therefore roots of $T^m - 1$. The m distinct powers of γ already give m roots; any further element would contradict the root bound for a polynomial over a field. Thus every nonzero element is a power of γ, and every element of F is a polynomial in γ over k_0. □

Proposition 7.4. *Suppose C is reduced with a finite basis $b_1, \ldots, b_n$ and nondegenerate trace. Suppose also that a finite subfield $k_0 \subseteq k$ contains the coefficients of monic annihilators of the b_i. Then C is a finite product of separable monogenic k-algebras.*

Proof. Let $A = k_0[b_1, \dots, b_n] \subseteq C$. The monic equations provide a finite spanning list of bounded monomials over k_0. Since k_0 is explicitly finite, the corresponding finite list of linear combinations enumerates A. Equality in C, and hence in A, is decidable from the given k-basis. The subalgebra A is reduced, so Lemma 7.2 supplies its finite idempotent decomposition.

For a nonzero piece eA, choose a primitive element γ by Lemma 7.3. The inclusion $eA \subseteq eC$ preserves the identity e. Each eb_i is a polynomial in γ over k_0, with coefficients interpreted as multiples of e. Since the eb_i span eC over k, this gives

$$eC = k[\gamma].$$

Lemma 4.2 supplies a finite basis and nondegenerate trace for eC, and reducedness is inherited from C. Its minimal polynomial is separable by Lemma 4.3. Thus $eC \simeq k[T]/(f_e)$ with f_e monic separable. If $eA = 0$, injectivity into C gives $e = 0$ and $eC = 0$, represented by $f_e = 1$.

The idempotents sum to 1 in A, hence also in C. The same projection and addition maps assemble $C \simeq \prod_e eC$, with the required monogenic presentations. □

8 Completion of the theorem

Proof of Theorem 1.2. Construct ε by Lemma 2.1. Proposition 2.3 makes C regular and reduced. Proposition 3.3 constructs a finite basis $b_1, \dots, b_n$, and Proposition 4.1 gives a perfect trace pairing. If $n = 0$, take the quotient by 1.

Otherwise choose monic annihilators of the b_i and collect their finitely many coefficients. Apply Lemma 7.1 with $M = n^3$. In the first alternative, Proposition 5.2 gives a tensor separator, and Proposition 6.2 gives a single separable monogenic presentation of C. In the second alternative, Proposition 7.4 gives the finite product. This proves the asserted decomposition.

For completeness, a monic separable quotient $k[T]/(f)$ is finite free and formally étale. To lift a root across a square-zero ideal N of an algebra B, choose a lift t of the root. The derivative $f'(t)$ is a unit, because it is a unit modulo N. The unique correction in N is

$$t \longmapsto t - f'(t)^{-1} f(t),$$

as $f(t+u) = f(t) + f'(t)u$ for $u \in N$. A unit modulo a square-zero ideal lifts to a unit by inverting $1+v$ as $1-v$ for $v \in N$. Finite products preserve this lifting property: their component idempotents lift uniquely across a square-zero ideal, after which one lifts each component map. Together with finite presentation, this proves that the product is finite étale. □

Remark 8.1. The small-field alternative is essential. In $\mathbb{F}_2^3$, every element satisfies $T^2 - T$, so the subalgebra generated by one element has dimension at most two. The product decomposition nevertheless has three monogenic factors. Over an arbitrary discrete field the argument supplies separable polynomials without requiring their factorisation into irreducibles. If such factorisations are supplied, the Chinese remainder theorem refines the result to a product of finite separable field extensions.

Part II
Connection with the formalisation

9 The precise result in Unit

The mathematical product theorem above is formalised in `unit/sag/etale-field-product`. Its endpoint is

```
UnramifiedFieldProduct.Over[k,setk,field]
  .At[C,fp,unique].theorem.
```

Here `field` supplies the zero-or-unit operation and $1 \neq 0$; `fp` is a finite presentation under propositional truncation; and `unique` is precisely uniqueness of square-zero lifting. Neither a basis nor a specified prime is an input.

The conclusion is the propositional truncation of `ProductPresentation.Over[k,setk].At[C]`. This record contains a finite binary tree of monic polynomials, an inverse of each derivative in the corresponding quotient, and an algebra isomorphism from the actual product to C. Both directions are algebra maps with proved inverse equations. The polynomials need not be irreducible, and zero factors are retained. The product is represented by its finite tree rather than flattened to one list of factors.

Existence statements in the mathematical argument are expressed through propositional truncation where appropriate. Finite witnesses are combined inside that truncation. There is no global choice of representatives or primitive elements. The development uses the existing univalence, replacement and surjection parameters of the shared theory and introduces no new axiom.

10 Correspondence of the proof steps

Mathematical step	**Principal formal components**
Diagonal and flatness transfer	`DiagonalCertificate`, `FlatDiagonalTransfer`, `UnramifiedFieldRegular`. These construct the diagonal and the averaging retraction from formal unramifiedness.
Finite span and basis	`ZeroDimensionalFinite`, `UnramifiedFieldFiniteGeneral`, `FinitePresentedFieldBasis`. Monic substitutions, integral descent, the finite monic cover and its relation matrix are constructed.
Trace duality	`TensorBasisConstruct`, `DiagonalTraceBridge`, `TraceDuality`, `FiniteUnramifiedTrace`. The tensor coordinates and their reconstruction are proved from the coproduct.
Minimal monic quotient	`MinimalPolynomial`, `MinimalPolynomialInjection`. The Gaussian search, injection and inverse algebra map when the generator is surjective are explicit.
Monogenic separability	`TraceDerivations`, `TraceRetraction`, `MonicTraceSeparable`, `GeneratedSeparableMonogenic`. The derivation killing f' is constructed using dual numbers.

Mathematical step	**Principal formal components**
Tensor separator	`DiagonalSeparator`, `FinitePolynomialSelection`, and the polynomial-basis and determinant modules. The scalar search has the explicit bound $n^3 + 1$.
Leading-coefficient functional	`FieldSeparatingFunctional`, `MinimalPowerFunctional`. These construct the quotient coordinate functional used in (4).
Divided difference and derivative inverse	`MonicDividedDifference`, `TensorDividedDifference`, `TensorMonicContraction`, `SeparatorDerivativeUnit`.
Global generation	`RegularUnitReflection`, `TensorPolynomialCoordinates`, `SeparatorMonogenic`, `LargeFieldMonogenic`. Preimages of the diagonal coordinates and then of every element are constructed.
Finite coefficient alternative	`FiniteMonicCoefficientFamily`, `FiniteCoefficientField`, `MonicFiniteImage`. The finite coefficient field is constructed, and the monic equations are lifted to it.
All finite idempotent pieces	`FiniteRegularRefinement`, `FiniteResolvedField`. Both branches of every split are retained; induction is on the enumeration length.
Primitive elements on the pieces	`FiniteFieldPrimitive`, `HornerCoefficientChange`, `PolynomialSpanGeneration`, `IdempotentMonogenicSpan`. Polynomial representatives span each ambient corner.
Product assembly	`IdempotentProductIsomorphism`, `ProductsOnIdempotents`, `SmallFieldProduct`, `FiniteEtaleProduct`, `UnramifiedFieldProduct`.

There are two useful distinctions between the mathematical presentation and its implementation. First, the determinant argument in Proposition 5.2 is implemented using matrix McCoy rather than constructing $k(Z)$ and clearing denominators. Polynomial McCoy supplies the preceding regularity statement. Second, trace nondegeneracy is passed to an idempotent factor using rectangular trace cyclicity and a retraction, without constructing a basis of the complementary factor. Both implementations establish exactly the lemmas used above.

The global-generator proof is the shorter argument used in the new product development. The earlier selected-prime development instead used relative bases on an idempotent neighbourhood of a given prime. Those earlier files remain available, but the product theorem has no prime parameter and does not use that selection step.

11 Validation and the remaining local connection

The recorded fresh check of the complete import tree gives 18757 declarations, 0 failures, in 9.42 seconds. Of these, 847 declarations in 25 owned modules are new. All 446 modules pass the module checks. The opening audit covers all 797 named new declarations and reports no openings outside their declared lists; each new declaration takes at most 0.03 seconds and opens at most 18 definitions in that run.

Regressions cover $\mathbb{F}_2$ from finite presentation and formal unramifiedness, the constructed standard algebra for $T^2 - T$, the large branch with its actual tensor separator, the small branch on $\mathbb{F}_2^3$, and the zero algebra. The $\mathbb{F}_2^m$ regression constructs its basis, trace pairing and monic equations and invokes the full small-field construction. It does not supply the desired product isomorphism. These checks validate proof terms; they do not extract and normalise the entire returned truncated

tree. Three isolated mutations—an incorrect divided difference, an incorrect complementary piece, and a missing derivative inverse—were rejected by the checker.

The formal endpoint is the explicit product isomorphism with derivative inverses. The final paragraph of the mathematical proof explains why these algebras are finite étale; the product milestone does not add a separate packaged theorem identifying its conclusion with every existing étale interface.

For the local structure problem over a general base, this result provides the full residue-field decomposition. Connecting it to the original local factor-selection proof still requires assembling the coefficient and denominator bridge on the exact special fibre and supplying the remaining local hypotheses. The existing selected-factor local adapter is unchanged. In particular, completion of this field theorem does not by itself assert completion of the unconditional local theorem that flat and unramified implies étale.

The implementation signatures, commands and validation details are recorded in `unit/sag/etale-field-product/README.txt`. The earlier note `etale-field-reduction.tex` also describes the selected-factor route and the local integration work; the present version separates the mathematical product proof from these formal connections.